

AIM2 Preparation

Week 1 - Day 3: Control Flow

5-Minute Review Notes | Conditions, Comparisons & Grade Logic

Day 3 Goal

Learn how Python makes decisions using conditions. Use if, elif, and else with comparison and logical operators to classify values and process student records.

1. What Is Control Flow?

Control flow determines which code runs based on conditions. Instead of executing every statement the same way, a program can choose different paths.

```
if condition:
    # runs when condition is True
```

2. if Statement

```
score = 85

if score >= 50:
    print("Pass")
```

The indented block executes only when the condition evaluates to True.

3. if / else

```
if score >= 50:
    print("Pass")
else:
    print("Fail")
```

else handles the alternative path when the if condition is False.

4. if / elif / else

```
if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
elif score >= 70:
    grade = "C"
elif score >= 60:
    grade = "D"
else:
    grade = "F"
```

Python checks conditions from top to bottom and stops at the first True condition. Therefore, the order of conditions matters.

5. Comparison Operators

Operator	Meaning	Example
==	Equal to	score == 85
!=	Not equal to	score != 0
>	Greater than	score > 80

Operator	Meaning	Example
<	Less than	score < 50
>=	Greater than or equal to	score >= 50
<=	Less than or equal to	score <= 100

6. Logical Operators

Operator	Meaning
and	Both conditions must be True.
or	At least one condition must be True.
not	Reverses a Boolean result.

```
if math >= 50 and python >= 50:
    print("Passed both subjects")
```

7. Conditions Produce Boolean Values

```
score = 85
print(score > 80) # True
print(score < 50) # False
```

A comparison itself evaluates to True or False. if uses that Boolean result to decide whether its block should run.

8. Grade Classification

```
average = 87

if average >= 90:
    grade = "A"
elif average >= 80:
    grade = "B"
elif average >= 70:
    grade = "C"
elif average >= 60:
    grade = "D"
else:
    grade = "F"
```

For 87, the >= 90 condition is False and >= 80 is True, so the result is grade B. No later branch is checked.

9. Processing Multiple Students

```
students = [
    {"Name": "Krishna", "Average": 88},
    {"Name": "Rahul", "Average": 74},
    {"Name": "Priya", "Average": 93}
]

for student in students:
    average = student["Average"]

    if average >= 90:
        grade = "A"
    elif average >= 80:
        grade = "B"
    else:
        grade = "C or below"
```

This combines Day 2 data structures with Day 3 control flow: loop through student records, read each score, and make a decision.

10. Indentation Matters

```
if score >= 50:
    print("Pass")
```

Python uses indentation to define the statements belonging to an if, elif, or else block. Incorrect indentation changes program structure or causes an error.

Day 3 Practice Highlights

- Used if, elif, and else to make decisions.
- Practiced comparison operators such as >=, <=, ==, and !=.
- Used logical operators to combine conditions.
- Built grade-letter logic from numerical scores.
- Processed a list of students and assigned results based on scores.
- Practiced reading conditions in top-to-bottom order.

Quick Syntax Cheat Sheet

```
if condition:
    ...
```

```
if condition:
    ...
else:
    ...
```

```
if condition1:
    ...
elif condition2:
    ...
else:
    ...
```

```
== != > < >= <=
and or not
```

Key Takeaways

- if runs a block when its condition is True.
- elif provides additional conditions; else handles the remaining case.
- Conditions are checked from top to bottom, so order matters.
- Comparison operators produce Boolean values.
- and, or, and not combine or modify conditions.
- Indentation defines Python control-flow blocks.
- Control flow lets raw numerical values become meaningful results such as grades and pass/fail classifications.

Progress

Week 1 Day 3 - Control Flow: COMPLETE

Practice focus: grading logic with if/elif/else and student records.